

SOFTWARE REQUIREMENTS SPECIFICATION

IntelliHire

AI-Powered Multilingual Voice Interview Preparation Platform

Document Type: SRS (IEEE 830-style)

Version: 1.1 — Final

Covers: Core Platform, Three-Panel System, Dynamic Interview Planning,
Interview Modes, Question Validation, Analytics, AI Explainability

Software Requirements Specification (SRS)

IntelliHire — AI-Powered Multilingual Voice Interview Preparation Platform

Document Version: 1.0 **Based on:** IntelliHire Architecture v3 – v6 (Enterprise Edition) **Prepared for:** Project submission / development reference

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for **IntelliHire**, an AI-powered, multilingual, voice-based interview preparation and assessment platform. It is intended for use by the development team as an implementation reference, and by evaluators/reviewers as a complete statement of what the system does, for whom, and under what constraints.

1.2 Scope

IntelliHire simulates realistic job interviews by combining resume understanding, adaptive AI-driven questioning, multilingual voice interaction, and automated performance evaluation. The system serves three distinct user roles — **Admin**, **Company (Recruiter)**, and **Candidate/Intern** — through three separate panels, and supports configurable interview behavior (Interview Modes) so that different hiring contexts (campus drives, senior technical hiring, HR/behavioral rounds) can be served by the same platform without code changes.

The system:

- Accepts a candidate's resume and extracts a structured knowledge profile from it.
- Dynamically plans an interview (topics, depth, and estimated question count) based on that profile.
- Conducts the interview by voice, in the candidate's chosen language, adapting question difficulty and topic flow to the candidate's live performance.
- Validates every AI-generated question before it reaches the candidate.
- Evaluates each answer across multiple dimensions, tracks resume-consistency and integrity signals, and produces a structured final report.
- Gives companies aggregate analytics across all their candidates.
- Gives the platform admin full control over what companies and candidates are permitted to configure.

Out of scope: payment/billing systems, native mobile applications (web-responsive only), video-based (facial/posture) analysis, and integration with third-party ATS (Applicant Tracking Systems) — these are not part of the current specification.

1.3 Definitions, Acronyms, Abbreviations

Term	Meaning
STT / TTS	Speech-to-Text / Text-to-Speech
LLM	Large Language Model
RBAC	Role-Based Access Control
Turn	One question–answer exchange within an interview session
Blueprint	The ordered, weighted list of topics an interview will cover, generated from the resume
Interview State	The live, continuously-updated object tracking everything about an in-progress interview
Interview Mode	A company-facing label (e.g., "Structured Interview") that maps internally to a question-asking strategy and a set of behavioral rules
Strategy (A/B/Hybrid)	Internal, non-company-facing prompt-construction techniques that control how questions are worded
Readiness Score	The platform's term for the candidate's measured performance/behavioral estimate (explicitly not called "Confidence Score" — see §3.6)
Guardrail	A deterministic filter that detects prompt-injection or exploit attempts in a candidate's spoken answer

1.4 References

This SRS consolidates and formalizes the decisions made across the IntelliHire Architecture documents (v3 "Groq/Gemini + no-RAG redesign," v4 "Interview Strategy Engine," v5 "Multi-Panel Platform + Strategy Framework," and the Enterprise Edition additions covering Interview Modes, Dynamic Interview Planning, the Question Validator Layer, and Company Analytics).

1.5 Overview

Section 2 describes the product at a high level. Section 3 lists detailed functional requirements grouped by subsystem. Section 4 covers external interfaces. Section 5 covers non-functional requirements. Section 6 covers data requirements. Section 7 covers other constraints.

2. Overall Description

2.1 Product Perspective

IntelliHire is a new, self-contained, three-tier web platform:

- **Presentation tier:** React.js + Tailwind CSS, serving three distinct panels (Admin, Company, Candidate) from role-based routing.

- **Application tier:** FastAPI (Python, async), hosting the Interview Manager, Interview Strategy Engine, LLM Service Layer, Question Validator, Analytics Service, and all REST/WebSocket APIs.
- **Data tier:** MongoDB (accessed via the async `motor` driver) as the sole persistent store. No relational database, no vector database, no object storage (S3/AWS) is used — see §2.5 for the rationale.

External dependencies: **Groq API** (primary LLM, `llama-3.3-70b-versatile`), **Gemini 2.5 Flash** (fallback LLM), and **Sarvam AI** (`saaras:v3` for STT, `bulbul:v3` for TTS).

2.2 Product Functions (Summary)

Function group	Description
Resume Intelligence	Parse, clean, and structure an uploaded resume into a JSON profile; discard the original file (no permanent storage)
Dynamic Interview Planning	Analyze the structured resume's experience/skill/project complexity to compute a <i>dynamic</i> question budget and generate a topic blueprint — no fixed question counts
Adaptive Interview Delivery	Ask voice questions via TTS, capture and transcribe voice answers via STT, adapt difficulty and topic flow turn-by-turn
Interview Mode Selection	Let a company pick a business-friendly "Interview Mode" (e.g., Structured, Conversational, Deep Technical, HR/Behavioral, Campus/Fresher, Balanced) that internally configures strategy, follow-up policy, and thresholds
Question Validation	Deterministically validate every AI-generated question before it is spoken to the candidate (duplicate check, relevance, difficulty alignment, clarity)
Evaluation & Scoring	Score every answer across technical, communication, and consistency dimensions; compute a Readiness Score; flag resume-consistency gaps and integrity risks
Security & Integrity	Detect and neutralize prompt-injection attempts (Guardrail Router); detect likely reading-from-script/pasted-AI-text answers (Anti-Cheat Detection)
Session Resilience	Recover an in-progress interview after a WebSocket disconnect, without re-running any LLM call
Reporting	Generate a structured final report per candidate (scores, strengths, weaknesses, risk assessment, recommendations)
Company Analytics	Aggregate metrics across a company's candidates — weak/strong skills, completion rates, strategy effectiveness, integrity incidents, etc.
Platform Administration	Manage companies, enable/disable strategies and interview modes platform-wide, view platform-wide analytics and integrity logs

2.3 User Classes and Characteristics

Role	Description	Typical actions
Admin	Platform owner/operator	Manage companies; enable/disable languages, voices, strategies, interview modes; view platform-wide analytics and security logs; configure global system settings

Role	Description	Typical actions
Company (Recruiter)	An organization using IntelliHire to screen its own candidates	Create Interview Campaigns; choose voice/language/accent, Interview Mode, and delegation flags; view their own candidates' reports and analytics dashboard
Candidate / Intern	The person being interviewed	Register (via a campaign-scoped invite), upload resume, optionally choose language/domain (if delegated by the company), take the voice interview, view their own report

2.4 Operating Environment

- **Client:** Any modern desktop or mobile browser supporting the [MediaRecorder](#) Web API (required for audio capture) and WebSocket connections.
- **Server:** Python 3.11+/FastAPI application, containerized (Docker), deployable to any standard cloud VM or container platform.
- **Database:** MongoDB (self-hosted via Docker for development; managed MongoDB Atlas or equivalent acceptable for production, at the implementer's discretion — no change to this SRS's data model either way).
- **External network access required to:** Groq API, Gemini API, Sarvam AI API. No other external network dependency exists (explicitly no AWS/S3 dependency).

2.5 Design and Implementation Constraints

- **No permanent file storage.** Uploaded resumes and answer audio are processed transiently and discarded; only derived structured data (resume JSON profile, answer transcripts) is persisted.
- **No vector database / RAG.** All per-turn context is delivered to the LLM via direct injection of a small, bounded Interview State object (resume profile + last 3 turns + blueprint/coverage metadata), not retrieval — justified by the small, single-purpose size of a resume knowledge base (see Architecture v3 §0 for the full trade-off discussion).
- **No Redis / external cache.** Session recovery and all live state reads use MongoDB directly, since `interview_state` is already a complete, bounded snapshot.
- **LLM provider must be provider-independent at the code level.** No caller outside the LLM Service Layer may import a provider SDK directly; provider selection and fallback (Groq → Gemini) are internal to that layer.
- **All topic/follow-up/completion/difficulty/question-count decisions must be deterministic Python, not LLM judgment,** except where content generation genuinely requires language understanding (resume structuring, blueprint generation, question wording, evaluation scoring, report writing). This is a firm architectural boundary carried through every version of the design.
- **Question Validator must be deterministic, not a second LLM call** — for cost, latency, and auditability reasons.

2.6 Assumptions and Dependencies

- Candidates have access to a microphone-enabled device and a stable-enough internet connection for real-time-adjacent (non-streaming) voice interaction.

- Groq, Gemini, and Sarvam AI API availability and rate limits are assumed sufficient for expected usage; the fallback provider and retry logic mitigate but do not eliminate dependency on these third parties.
- Resumes are assumed to be in PDF or DOCX format and predominantly text-based (not scanned images) for reliable parsing.

3. Functional Requirements

Each requirement is tagged with an ID for traceability (`FR-<module>-<number>`).

3.1 Authentication & Role-Based Access Control

ID	Requirement
FR-AUTH-1	The system shall authenticate all users via JWT, issued at login/registration.
FR-AUTH-2	Each JWT shall encode <code>role</code> (<code>admin</code> <code>company</code> <code>candidate</code>), and where applicable <code>company_id</code> and <code>campaign_id</code> .
FR-AUTH-3	The system shall enforce role-based access on every API endpoint via a centralized permission dependency; no endpoint shall rely on client-supplied role information.
FR-AUTH-4	Company-scoped and candidate-scoped data access shall be filtered server-side using the <code>company_id</code> / <code>campaign_id</code> from the verified token, never from request payload fields.
FR-AUTH-5	A company's available configuration choices (languages, voices, strategies, interview modes) shall be restricted server-side to the subset the Admin has enabled for that company (<code>companies.allowed_*</code> fields).
FR-AUTH-6	A candidate's available choices (language, domain) shall be restricted server-side to the subset the company has delegated for that specific campaign.

3.2 Resume Processing

ID	Requirement
FR-RES-1	The system shall accept resume uploads in PDF or DOCX format.
FR-RES-2	The system shall extract raw text using PyMuPDF/pdfplumber (PDF) or python-docx (DOCX).
FR-RES-3	The system shall clean extracted text (remove headers/footers/page numbers, normalize whitespace, repair line-wrap breaks) before further processing.
FR-RES-4	The system shall use the LLM Service to structure the cleaned resume text into a JSON profile containing: candidate name, target role, experience, education, skills, projects, certifications, strengths, technologies, domains, and potential interview topics.

ID	Requirement
FR-RES-5	The system shall persist only the structured JSON profile in MongoDB; the original uploaded file shall not be written to any persistent or object storage and shall be discarded immediately after structuring completes.
FR-RES-6	If resume structuring fails after all LLM Service retries/fallback, the system shall surface a clear error to the candidate and shall not create an interview session from an incomplete profile.

3.3 Dynamic Interview Planning

ID	Requirement
FR-PLAN-1	Immediately after resume structuring, the system shall run an Experience Analyzer that estimates a numeric experience-complexity score from years of experience, number/seniority of past roles, internships, and research history.
FR-PLAN-2	The system shall run a Skill Complexity Analyzer that scores the candidate's primary and secondary skills by breadth and typical interview depth required.
FR-PLAN-3	The system shall run a Project Complexity Analyzer that scores each listed project by technical scope (e.g., number of technologies, system complexity indicators, described impact).
FR-PLAN-4	The system shall compute a dynamic question budget (a <code>[min_questions, max_questions]</code> range) as a weighted function of: experience score, project complexity score, primary-skill count, secondary-skill count, selected difficulty policy, selected Interview Mode, and target role — replacing any fixed, hardcoded question-count constants.
FR-PLAN-5	The computed question budget shall fall within platform-defined sanity bounds (absolute floor and ceiling) regardless of the weighted score, to prevent degenerate (too-short or unbounded) interviews.
FR-PLAN-6	The Dynamic Interview Planner's output (question budget + complexity scores) shall be passed to the Interview Blueprint Generator and stored in Interview State before the first question is issued.
FR-PLAN-7	The Completion Engine's Maximum Question Guard shall use the dynamically computed <code>max_questions</code> for that specific session, not a platform-wide constant.

3.4 Interview Blueprint & Topic Management

ID	Requirement
FR-BP-1	The system shall generate an ordered Interview Blueprint (list of topics, each tagged with section, importance, priority rank, target difficulty, and estimated coverage depth) once per session, using the LLM Service, informed by the resume profile and the Dynamic Interview Planner's output.
FR-BP-2	The system shall maintain a live Topic Coverage Map, updated after every turn, tracking per-topic status (not started / in progress / covered / failed-abandoned), questions asked, follow-ups asked, and a recency-weighted topic confidence score.
FR-BP-3	The Topic Priority Manager shall always select mandatory topics before high/medium/low-importance topics when choosing the next topic.
FR-BP-4	The system shall never re-ask a topic already marked "covered" or "failed-abandoned" except as an explicit resume-consistency re-check (see §3.9).

3.5 Interview Mode & Strategy Selection

ID	Requirement
FR-MODE-1	The Company Panel shall expose only business-facing Interview Modes — Structured, Conversational, Balanced (default), Deep Technical, HR & Behavioral, and Campus/Fresher — and shall never expose the internal strategy names (Strategy A/B/Hybrid) to company users.
FR-MODE-2	The Interview Mode Manager shall map each selected Interview Mode to: an internal strategy (A, B, or Hybrid), a follow-up policy (max follow-ups per topic, aggressiveness), a topic-confidence/saturation threshold, a difficulty policy (starting difficulty, progression aggressiveness), and — for HR & Behavioral mode — a behavioral prompt template set (STAR framework, leadership, conflict resolution, communication, decision-making, teamwork).
FR-MODE-3	Structured Interview mode shall configure: Strategy A, minimal transition text, low follow-up count ceiling, and a topic-coverage-first prioritization suited to high-volume/campus hiring.
FR-MODE-4	Conversational Interview mode shall configure: Strategy B, full transition/bridging text, relaxed tone instructions in the prompt template.
FR-MODE-5	Balanced Interview mode (platform default) shall configure: Hybrid strategy, natural transitions, and moderate follow-up/coverage settings — selected as default because it provides both topic-drift protection (from Strategy A) and natural conversational quality (from Strategy B) without the trade-offs of either extreme.
FR-MODE-6	Deep Technical Interview mode shall configure: Hybrid strategy, an elevated topic-confidence/saturation threshold (i.e., harder to mark a topic "covered"), a higher follow-up ceiling per topic, and prompt instructions biased toward "why," architecture, and design-tradeoff questions.
FR-MODE-7	HR & Behavioral Interview mode shall configure: automatic switching to behavioral question templates (STAR-format prompts) for the relevant blueprint sections, independent of the candidate's technical resume content.
FR-MODE-8	Campus/Fresher Interview mode shall configure: a lower starting difficulty, a shallower follow-up depth, and prompt instructions favoring encouraging phrasing and resume-anchored (rather than open-ended) question framing.
FR-MODE-9	All Interview Mode → internal-parameter mappings shall be stored as configuration data (not hardcoded conditionals), allowing new modes to be added by the Admin without a code deployment.

3.6 LLM Service Layer

ID	Requirement
FR-LLM-1	All LLM calls in the system shall route through a single LLM Service Layer; no other module shall import or call a provider SDK directly.
FR-LLM-2	The LLM Service shall attempt the primary provider (Groq, llama-3.3-70b-versatile) first, with automatic fallback to the secondary provider (Gemini 2.5 Flash) on failure or invalid response, after a bounded number of retries per provider.
FR-LLM-3	The LLM Service shall validate every LLM response against an explicit schema before returning it to the caller; invalid/unparseable responses shall trigger a retry or fallback, not be passed through.

ID	Requirement
FR-LLM-4	Prompt construction shall be handled entirely within the LLM Service (via versioned prompt templates), never inline within business-logic modules.
FR-LLM-5	The LLM Service shall support distinct, addressable operations: resume structuring, blueprint generation, per-turn evaluation-and-next-question generation, resume consistency checking, and final report generation.

3.7 Voice Interaction

ID	Requirement
FR-VOICE-1	The system shall convert every AI-generated question to speech using Sarvam AI <code>bulbul:v3</code> , in the language and voice configured for the active campaign.
FR-VOICE-2	The frontend shall enable the "Start Answering" control only after question audio playback completes.
FR-VOICE-3	The system shall capture the candidate's spoken answer via the browser <code>MediaRecorder</code> API, beginning on "Start Answering" and ending on "Submit Answer."
FR-VOICE-4	The system shall transcribe submitted audio using Sarvam AI <code>saaras:v3</code> ; the resulting transcript shall be treated as the candidate's official answer for all downstream evaluation.
FR-VOICE-5	Recorded audio shall not be persisted beyond the transcription step (consistent with the no-permanent-audio-storage constraint, §2.5).
FR-VOICE-6	The system shall not require real-time interruption or streaming evaluation; evaluation begins only after the candidate explicitly submits an answer.

3.8 Question Validation Layer

ID	Requirement
FR-QV-1	Every question generated by the LLM Service shall pass through a deterministic Question Validator before being sent to Text-to-Speech and shown/spoken to the candidate.
FR-QV-2	The Validator shall check for duplicate questions using a combination of keyword overlap and semantic/embedding similarity against the current session's already-asked questions, rejecting any candidate question above a configured similarity threshold.
FR-QV-3	The Validator shall check topic relevance , confirming the question addresses the currently selected blueprint topic (or a valid follow-up thereof) and not an unrelated subject.
FR-QV-4	The Validator shall check difficulty alignment , confirming the question's apparent difficulty is consistent with the value set by the Difficulty Controller for that turn.
FR-QV-5	The Validator shall check resume/blueprint alignment , rejecting questions that reference skills, projects, or claims not present in the candidate's structured resume profile or blueprint (to prevent hallucinated references).
FR-QV-6	The Validator shall check that a proposed follow-up question is a valid follow-up (i.e., logically connected to the prior answer) rather than a disguised topic change.

ID	Requirement
FR-QV-7	The Validator shall check basic question clarity and length (non-empty, within a reasonable word-count range, syntactically well-formed) before acceptance.
FR-QV-8	The Validator shall check Interview Mode compliance (e.g., a Structured-mode session shall not receive an overly conversational, multi-clause question).
FR-QV-9	On validation failure, the system shall automatically request regeneration from the LLM Service, up to a configured maximum retry count.
FR-QV-10	If the maximum retry count is exhausted, the system shall apply a defined fallback policy (e.g., serve a pre-approved template question for the current topic/difficulty) rather than blocking the interview.
FR-QV-11	Every validation attempt (pass/fail, failed rule(s), retry count) shall be logged for later analytics and debugging.

3.9 Evaluation, Difficulty, and Consistency

ID	Requirement
FR-EVAL-1	The system shall evaluate every submitted answer across: technical knowledge, communication, completeness, logical flow, resume consistency, project explanation quality, professionalism, and response quality — via a single LLM Service call per turn.
FR-EVAL-2	Response time and topic coverage contribution shall be computed in Python from timestamps and state, not requested from the LLM.
FR-EVAL-3	The system shall compute an Interview Readiness Score (not labeled "Confidence Score") as a deterministic weighted formula over technical correctness, communication quality, completeness, resume consistency, response quality, response time, and topic coverage — never as a value directly asked of the LLM.
FR-EVAL-4	The Difficulty Controller shall increase, hold, or decrease difficulty based on the latest score, and shall additionally apply a CALIBRATE_HOLD (Adaptive Pace Adjuster) when a high score is paired with an unusually long response time, to avoid escalating difficulty on a cognitively strained candidate.
FR-EVAL-5	The Follow-up Decision Engine shall decide, per turn, whether to ask a follow-up on the same topic or move to the next topic, based on topic coverage state, the evaluation's suggests_follow_up signal, and the active Interview Mode's follow-up policy — not a fixed per-topic question count.
FR-EVAL-6	The Resume Consistency Checker shall flag a "Resume Skill Gap" when a candidate's demonstrated performance on a topic contradicts a corresponding resume claim (e.g., resume lists a skill, but scored performance on that topic is low).

3.10 Interview Completion

ID	Requirement
FR-COMP-1	The system shall terminate an interview only when at least one of the following holds: (a) the dynamically computed maximum question count is reached (Emergency Exit), or (b) all mandatory blueprint topics are marked "covered" and overall interview confidence meets the active Interview Mode's threshold (Clean Completion).

ID	Requirement
FR-COMP-2	The system shall never terminate below the dynamically computed minimum question count, regardless of early high performance.
FR-COMP-3	A topic shall be marked "covered" (saturated) only once its confidence score is both high and stable across at least two answers on that topic, not from a single high-scoring answer.
FR-COMP-4	A topic on which the candidate scores below a low-performance threshold on two consecutive attempts shall be marked "failed-abandoned" and excluded from further selection, to avoid prolonging candidate distress on a clearly weak area.
FR-COMP-5	If the interview terminates via Emergency Exit with mandatory topics still incomplete, the final report shall be explicitly flagged <code>incomplete_coverage: true</code> .

3.11 Security & Integrity

ID	Requirement
FR-SEC-1	Every submitted transcript shall be checked by the Guardrail Router (deterministic pattern matching, not an LLM call) for prompt-injection or exploit attempts before evaluation.
FR-SEC-2	On Guardrail detection, the system shall skip evaluation for that turn, not advance topic/coverage state, increment an exploitation-attempt counter on the session, and respond with a neutral boilerplate redirect spoken to the candidate.
FR-SEC-3	Every submitted transcript shall be checked by Anti-Cheat Detection (deterministic pacing/structure heuristics) for signals consistent with reading a script or pasted AI-generated text.
FR-SEC-4	Anti-cheat flags shall be recorded on the turn and surfaced only in the company/recruiter-facing report, never disclosed to the candidate during the interview.

3.12 Session Resilience

ID	Requirement
FR-SESS-1	On WebSocket disconnect, the system shall preserve the in-progress session rather than terminating or discarding it.
FR-SESS-2	On reconnect within a defined recovery window (5 minutes) with a valid JWT for that session, the system shall restore the candidate to their exact last question without re-invoking any LLM call.
FR-SESS-3	Session recovery shall use MongoDB's persisted <code>interview_state</code> as the sole source of truth; no additional caching layer (e.g., Redis) shall be introduced for this purpose.

3.13 Reporting

ID	Requirement
FR-REP-1	On completion, the system shall precompute all quantitative aggregates (per-topic averages, difficulty progression, score trends) in Python — never delegated to the LLM for calculation.

ID	Requirement
FR-REP-2	The system shall generate qualitative report sections (strengths, weaknesses, improvement plan, learning resources, recruiter-style summary) via a single LLM Service call, informed by the precomputed aggregates, full interview transcript, and resume profile.
FR-REP-3	The final report shall include: Overall Score, Interview Readiness Score, Resume Match Analysis, Topic-wise Scores, Technical Skills Assessment, Communication Assessment, Interview Risk Assessment, Strengths, Weaknesses, Personalized Improvement Plan, Learning Resources, and a Recruiter-style Summary.
FR-REP-4	Candidates shall be able to view their own report only; companies shall be able to view all reports belonging to their own campaigns only; Admin shall be able to view all reports platform-wide.

3.13A AI Interview Explainability

ID	Requirement
FR-EXP-1	On completion, the system shall generate a structured explanation of the interview's own decisions: why each topic was selected, why difficulty changed at each point, why each follow-up question was asked, why the interview ended, and how the Readiness Score was calculated.
FR-EXP-2	Explanations shall be generated deterministically from data already computed and stored during the interview (blueprint importance tags, score history, <code>suggests_follow_up</code> / <code>follow_up_reason</code> fields, Completion Engine trigger reason, Readiness Score formula) — the Explainability Generator shall not invoke the LLM and shall not introduce any new evaluative judgment.
FR-EXP-3	The explanation shall be persisted as a nested field on the same <code>interview_reports</code> document as the rest of the final report and served through the existing report retrieval endpoint — no separate storage or endpoint shall be introduced.
FR-EXP-4	The explanation shall be presented to the candidate and to the company/recruiter as a distinct, collapsible section of the report, positioned after the scored sections.

3.14 Company Analytics Dashboard

ID	Requirement
FR-AN-1	The Company Panel shall provide an aggregate analytics dashboard scoped to that company's own campaigns and candidates only.
FR-AN-2	The dashboard shall report: most common weak skills, strongest skills, average interview score, average interview duration, average Readiness Score, difficulty distribution, interview completion rate, drop-off rate, question-count distribution, per-strategy effectiveness, per-Interview-Mode effectiveness, average follow-up count, average topic coverage percentage, cheating-incident count, guardrail-trigger count, resume-consistency statistics, top failed topics, most difficult topics (by average score), and most successful Interview Mode (by completion rate and average score).
FR-AN-3	All dashboard metrics shall be computed via MongoDB aggregation pipelines over <code>interview_sessions</code> and <code>interview_reports</code> , filtered by <code>company_id</code> and an optional <code>campaign_id</code> .
FR-AN-4	The dashboard shall support filtering by a configurable time range.

ID	Requirement
FR-AN-5	The dashboard shall support exporting the current view as CSV and as PDF.
FR-AN-6	The Admin Panel shall provide the equivalent aggregate view without company scoping, i.e., platform-wide.

3.15 Administration

ID	Requirement
FR-ADM-1	Admin shall be able to create, suspend, and configure companies, including each company's <code>allowed_languages</code> , <code>allowed_voices</code> , <code>allowed_strategies</code> , <code>allowed_interview_modes</code> , and <code>allowed_llm_tiers</code> .
FR-ADM-2	Admin shall be able to enable or disable individual strategies and Interview Modes platform-wide.
FR-ADM-3	Admin shall be able to view platform-wide analytics, including guardrail-trigger and anti-cheat-flag logs across all companies.

4. External Interface Requirements

4.1 User Interfaces

- Responsive React + Tailwind web application with three role-scoped panels (Admin, Company, Candidate), served from the same frontend codebase with role-based routing.
- Candidate interview screen shall present: current question (audio + optional text), a clearly disabled/enabled "Start Answering" control gated on TTS playback completion, a recording indicator, and a "Submit Answer" control.

4.2 API Interfaces

- REST endpoints (JSON over HTTPS) for all CRUD-style operations across all three panels (see Architecture v5/v6 §9 for the full endpoint table).
- One WebSocket endpoint per active interview session (`/ws/interview/{session_id}`) for server-push events: TTS-ready, evaluation-in-progress status, next-question-ready, session-recovered.

4.3 External Service Interfaces

Service	Interface	Purpose
Groq API	REST, <code>llama-3.3-70b-versatile</code>	Primary LLM for all generation/evaluation tasks
Gemini API	REST, <code>gemini-2.5-flash</code>	Fallback LLM
Sarvam AI	REST, <code>saaras:v3</code> / <code>bulbul:v3</code>	STT / TTS

4.4 Hardware Interfaces

- Client device microphone, accessed via the browser's `MediaRecorder` Web API. No other hardware interface is required.
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5. Non-Functional Requirements

5.1 Performance

- Per-turn round trip (STT → evaluation/next-question LLM call → validation → TTS) shall complete within a few seconds under normal provider latency, acceptable given the explicit no-real-time-streaming design decision.
- The dynamic question budget and blueprint generation (one-time, at session start) may take longer than a per-turn call, and shall show a loading/setup state to the candidate rather than a blocked UI.

5.2 Reliability & Availability

- LLM calls shall automatically fail over from Groq to Gemini on error, per FR-LLM-2, so a single provider outage does not halt active interviews.
- Sessions shall survive transient WebSocket disconnects without data loss, per FR-SESS-1–3.

5.3 Security

- All endpoints shall enforce RBAC server-side (§3.1); no client-side-only permission checks.
- No permanent storage of raw resumes or raw audio (§2.5) reduces data-retention risk surface by design.
- Guardrail and anti-cheat detections (§3.11) run on every submitted transcript without exception.

5.4 Scalability

- The FastAPI application shall be stateless per request (all state in MongoDB), enabling horizontal scaling behind a load balancer without session affinity.
- Strategies and Interview Modes are configuration data, not code, allowing new ones to be added without a deployment (§3.5, FR-MODE-9).

5.5 Usability

- Company users shall never be exposed to implementation-level terminology (Strategy A/B/Hybrid); all company-facing configuration uses business-friendly Interview Mode names (§3.5).
- Candidates shall receive clear, unambiguous audio prompts for every interaction step (listen → start answering → submit).

5.6 Maintainability

- All LLM-provider-specific code shall be isolated to the LLM Service Layer's provider modules (§3.6), so provider changes do not ripple into business logic.
 - All deterministic decision logic (difficulty, completion, validation, coverage) shall remain in plain, testable Python functions, separated from LLM-dependent code paths.
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6. Data Requirements (Summary)

Primary MongoDB collections: `users`, `companies`, `interview_campaigns`, `strategy_definitions`, `interview_mode_definitions`, `candidates`, `interview_sessions` (embedding `interview_state` and `turns[]`), `interview_reports`, `validator_logs`, `analytics_aggregates` (or computed on-demand via aggregation pipelines rather than materialized, at the implementer's discretion).

Full field-level schemas for each collection are defined in the corresponding Architecture documents (v3 §3, v4 §3/§6.1, v5 §3, and the Enterprise Edition's Part 7 MongoDB updates) and are treated as authoritative for implementation; this SRS defines *what* must be stored, those documents define the exact shape.

7. Other Requirements

7.1 Legal / Compliance Considerations

- Since resumes and audio are not permanently stored, standard data-minimization practice is inherently supported; however, transcripts and structured resume profiles **are** retained and should be covered by an appropriate privacy notice to candidates at upload/registration time (implementation detail outside this SRS's technical scope, but flagged as a requirement on the product/legal side).

7.2 Assumptions on Terminology

- "Confidence Score" is deliberately never used in any candidate- or company-facing output; the platform term is **Interview Readiness Score**, computed from measurable behavioral and performance signals, not inferred emotion or visual cues.
- "Risk Analysis" is deliberately termed **Interview Risk Assessment** throughout candidate- and company-facing surfaces.

7.3 Traceability

Every functional requirement in Section 3 maps directly to a named component in the Architecture documents (Interview Manager, Interview Strategy Engine, LLM Service, Difficulty Controller, Resume Consistency Checker, Question Validator, Interview Mode Manager, Dynamic Interview Planner, Guardrail

Router, Anti-Cheat Detection, Analytics Service) — this SRS does not introduce any component not already defined architecturally; it formalizes their required behavior as testable requirements.